10.10 - Alternating Series Error Bound

Learning Objective(s):
LIM-7.B: Approximate the sum of a series.

Alternating Series Remainder

THEOREM 10.10: ALTERNATING SERIES REMAINDER

If a convergent alternating series satisfies the condition $a_{n+1} \le a_n$, then the absolute value of the remainder R_n involved in approximating the sum S by S_n is less than (or equal to) the first neglected term. That is,

$$|S - S_n| = |R_n| \le |a_{n+1}|$$

Example 1: Approximating the Sum of an Alternating Series
Approximate the sum of the following series by its first six terms.

$$= 1 - \frac{1}{e} = 0.6321206$$

$$S = \sum_{n=1}^{\infty} (-1)^{n+1}\left(\frac{1}{n!}\right) = \frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \frac{1}{5!} - \frac{1}{6!} + \cdots \approx S_6$$

$$= 0.6319444$$

$$|R_6| = |S - S_6| \le \frac{1}{7!} = 1.984 \times 10^{-4}$$

$$[0.6317460, 0.6321429]$$

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Example 2: When Does an Alternating Series Provide a Sum to a Given Accuracy?
Determine the number of terms required to approximate the sum of the given convergent series with an error of less than 0.001.

a.) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \sum_{n=1}^{\infty} (-1)^{n+1} a_n$

$$a_n = \frac{1}{n^2}$$

$$|R_n| \le |a_{n+1}| = \frac{1}{(n+1)^2} < \frac{1}{1000}$$

$$(n+1)^2 > 1000$$

$$n \ge 31$$

b.) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!}$

$$a_n = \frac{1}{2^n n!}$$

$$|R_n| \le |a_{n+1}| = \frac{1}{2^{n+1}(n+1)!} < 0.001$$

$$2^{n+1} \cdot (n+1)! > 1000$$

$$n \ge 4$$

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10.11 - Finding Taylor Polynomial Approximations of Functions

Learning Objective(s):

LIM-8.A: Represent a function at a point as a Taylor polynomial.

LIM-8.B: Approximate function values using a Taylor polynomial.

For over 300 years, mathematicians have been able to compute transcendental values with an alarming degree of accuracy. For example, how do you think a 17th century mathematician would be able to find (i) sin(0.2), (ii) ln(1.05), or (iii) $\sqrt[4]{e}$ without the convenience of a modern calculator?

Linear and Quadratic Approximations

Back in Topic 4.6 (from Calculus AB), we learned that if a function is differentiable at a point c then it can be approximated near c by its tangent line, which we called the linear approximation to f at the point c. Furthermore, we rewrote the point-slope version of that line, $y - f(c) = f'(c)(x-c)$ as

$$L(x) = f(c) + f'(c)(x-c).$$

Because this linear approximation is a first-degree polynomial of x, we can name it $P_1(x)$:

$$P_1(x) = f(c) + f'(c)(x-c).$$

This polynomial has some important properties.

1.) It matches the function f in value at $x = c$.

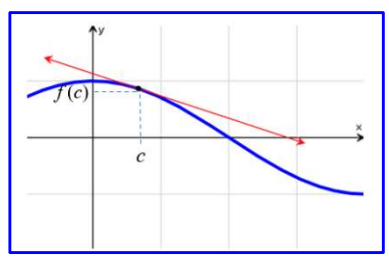
$P_1(c) = f(c) + f'(c)(c-c) = f(c)$

2.) It matches the slope of the function f in value at $x = c$.

$P_1'(c) = 0 + f'(c)(1) = f'(c)$

The problem is that linear approximations don't always work so well when the graph of f has a great deal of "curvature" near c.

To fix this problem, we can create a quadratic approximating polynomial by adding one new term to the linear polynomial. We can denote the new polynomial $P_2(x)$ and it would look like



$P_2(x) = f(c) + f'(c)(x-c) + a(x-c)^2$

Notice that we will need to find the value of the coefficient a. To determine a and to ensure that $P_2(x)$ is a good approximation for f near the point c we require that $P_2(x)$ agree with f in value, slope and concavity at c. In other words, $P_2(x)$ must satisfy the conditions

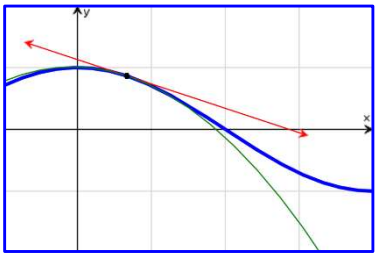
$P_2'(x) = f'(c) + 2a(x-c)$

$P_2''(x) = 2a = f''(c)$

$P_2(c) = f(c) + f'(c)(c-c) + a(c-c)^2 = f(c)$

$P_2'(x) = 0 + f'(c)(1) + 2a(x-c) \rightarrow P_2'(c) = f'(c) + 2a(c-c) = f'(c)$

$P_2''(x) = 2a \rightarrow P_2''(c) = 2a$ must be equivalent to $f''(c) \rightarrow a = \frac{1}{2}f''(c)$



Therefore, the resulting quadratic approximating polynomial is

$$P_2(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2}(x-c)^2$$

<u>Example 1</u>: Linear and Quadratic Approximations for ln(*x*)

Note: A calculator is only suggested for this problem to hasten the mundane arithmetic in part c.

a. Find the linear approximation to $f(x) = \ln x$ at $x = 1$.

$f(1) = 0$ $\quad f'(x) = \frac{1}{x} = x^{-1}$

$f'(1) = 1$

$P_1(x) = 0 + 1(x-1)$

$= x - 1$

b. Find the quadratic approximation to $f(x) = \ln x$ at $x = 1$.

$f''(x) = -1x^{-2}$

$f''(1) = -1$

$P_2(x) = (x-1) - \frac{(x-1)^2}{2}$

c. Use these approximations to estimate the value of $\ln(1.05)$.

$\ln(1.05) \approx P_1(1.05) = 0.05$

$\ln(1.05) \approx P_2(1.05) = 0.05 - \frac{0.05^2}{2} = 0.04875$

How do these approximations compare to the actual value of ln(1.05)?

$\ln(1.05) = 0.04879$

$|f(1.05) - P_1(1.05)| = 0.00121$

$|f(1.05) - P_2(1.05)| = 0.00004$

The big question now is whether or not we can extend the idea of linear and quadratic polynomials that will approximate functions to higher-degree polynomials that can perhaps to a better job of approximating functions.

Fortunately, the answer is YES.

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Taylor Polynomials

Observe $P_1(x) = f(c) + f'(c)(x-c)$ and

$$P_2(x) = f(c) + f'(c)(x-c) + a(x-c)^2.$$

Therefore, it is only logical that

$$P_3(x) = f(c) + f'(c)(x-c) + a(x-c)^2 + b(x-c)^3$$

and that it remains important to satisfy the four conditions

$$P_3(c) = f(c), \quad P_3'(c) = f'(c), \quad P_3'' = f''(c), \text{ and } P_3'''(c) = f'''(c).$$

Because $P_3(x)$ is built using $P_2(x)$, the first three conditions above are met.

The last condition, $P_3'''(c) = f'''(c)$, is used to find the value of the coefficient b.

By the time we reach the third derivative of $P_3(x)$, we notice

$P_3'(x) = f'(c) + 2a(x-c) + 3b(x-c)^2$

$P_3''(x) = 2a + 3 \cdot 2 \cdot b(x-c)$

$P_3'''(x) = 3 \cdot 2 \cdot b$ which is equivlanet to $3! \cdot b$

So, $P_3'''(c) = 3!(b)$ which must be equivalent to $f'''(c) \rightarrow \quad b = \dfrac{f'''(c)}{3!}$

MATHEMATICS & HISTORY

BROOK TAYLOR (1685-1731)
An English mathematician, Brook Taylor is best known for his contributions to the study of infinite series. Although he did not live to be very old, he made many contributions to the world of mathematics and is credited with the discovery of integration by parts.

With these coefficients, we can obtain the following definition of **Taylor polynomials**, named after the English mathematician Brook Taylor (1685-1731), and **Maclaurin polynomials**, named after the English mathematician, Colin Maclaurin (1698-1746).

DEFINITION OF *n*th TAYLOR POLYNOMIAL AND *n*th MACLAURIN POLYNOMIAL

If f has n derivatives at c, then the polynomial

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is called the ***n*th Taylor polynomial for *f* centered at *c***.

If c = 0, then

$$P_n(x) = f(0) + f'(0)(x) + \frac{f''(0)}{2!}(x)^2 + \cdots + \frac{f^{(n)}(0)}{n!}(x)^n$$

is also called the ***n*th Maclaurin polynomial for *f***.

Example 2: Maclaurin Polynomial for sin(x)

a. Find the seventh degree Maclaurin polynomial, $P_7(x)$, for $f(x) = \sin x$.

$f'(x) = \cos x = f^{(3)}(x)$

$f''(x) = -\sin x = f^{(5)}(x)$

$f'''(x) = -\cos x = f^{(7)}(x)$

$f^{(4)}(x) = \sin x = f(x)$

$f'(0) = f^{(6)}(0)$

$= f(0) = f^{(4)}(0) = 0$

$f'(0) = 1 = f^{(2)}(0)$

$f'''(0) = -1 = f^{(7)}(0)$

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b. Use $P_7(x)$ to approximate $\sin(0.2)$.

$P_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$

$\sin(0.2) \approx P_7(0.2) = 0.1986693308$

c. What do you notice about the degree of each term of $P_7(x)$? Can you make an assumption about the degree of any Maclaurin polynomial for $\cos x$?

Even!

Example 3: Writing a Taylor Polynomial Using Information Using a Table of Values

A function $f(x)$ is differentiable for all orders n and has the following derivative values when $x = 1$.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
1	2	−3	1	−6

Find a third-degree Taylor polynomial for $f(x)$ centered at $x = 1$.

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

$P_3(x) = 2 - 3(x-1) + \frac{1}{2}(x-1)^2 - (x-1)^3$

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Example 4: Writing a Maclaurin Polynomial Using Known Information

A function $g(x)$ has derivatives of all orders at $x = 0$. It is known that $g(0) = 2$, $g'(0) = -1$ and $g^{(n)}(0) = \frac{n^2}{n!}$ for $n \geq 2$. Find a third-degree Maclaurin polynomial for $g(x)$.

$g''(0) = \frac{2^2}{2!} = 2$

$g'''(0) = \frac{3^2}{3!} = \frac{9}{6} = \frac{3}{2}$

$P_3(x) = 2 - x + x^2 + \frac{1}{4}x^3$

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One of the primary goals of studying Taylor Polynomials and, later in Topic 10.14, Taylor Series is to find a way to represent an "ugly" function as a "prettier" function that we can manipulate much easier.

Ugly Functions	**Pretty (Nicer) Functions**
e^x	$1 + x + \frac{x^2}{2}$
$\ln(x)$	$(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$
$\sin(x)$	$x - \frac{x^3}{3!} + \frac{x^5}{5!}$
$\cos(x)$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!}$
$\tan^{-1}(x)$	$x - \frac{x^3}{3} + \frac{x^5}{5}$